

When Accepted Answers Become Evidence: Feedback Loops in Self-Updating Retrieval

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Abstract

Support systems built on retrieval often store questions whose answers users accepted, linked to the thread that answered them, so that the next similar question can be answered from the resolved case. This is the retain step of case-based reasoning, and a change to the retrieval half of a retrieval-augmented generation (RAG) pipeline; no text is generated. We measure what this human-gated loop does on real data: all twelve CQADupStack forums (446,641 posts, 23,703 questions), community duplicate links as ground truth, real repeat questions replayed through three retrievers (bge-small-en-v1.5, all-MiniLM-L6-v2, BM25). Only user acceptance is simulated. (1) When the stored entry is the question text, write-back with dense retrievers helps while users reject most wrong answers and hurts beyond a crossover acceptance rate p^* of wrong answers that ranges from below 0.10 to about 0.86 across configurations; at $p_w = 0.95$ the loss is significant in 22 of 24 dense configurations. p^* is a property of a configuration, not a general constant. (2) The loss comes from question-to-question similarity: a stored wrong question outscores the correct post. Storing the question together with the shown thread’s text removes most of the loss and most of the gain. (3) A plug-in estimate of p^* from an exact per-entry decomposition, calibrated on five orders and tested on five others, has mean absolute error 0.085 for question entries. (4) Version 3 adds two baselines from neighbouring literatures. A Doc2Query-- style filter that drops (question, thread) pairs with a low cross-encoder score before write-back halves the mean loss at $p_w = 0.95$ for question entries (-1.23 to -0.59 points, dense retrievers) but does less than reopening wrong answers (-0.36). Serving stored questions as a semantic cache is harmful at loose similarity thresholds for all three retrievers, including BM25, which is nearly unaffected by ordinary write-back; strict thresholds make the cache nearly inert. (5) On a second, heterogeneous corpus (MS MARCO passages, 2,500 families of distinct real Bing queries sharing a relevant passage) the same pattern appears with higher crossovers ($p^* = 0.58$ for bge, 0.78 for MiniLM). Label-free monitors add little to the acceptance rate.

1 Introduction

A common design in customer support and IT service management is to treat resolved cases as new knowledge: when a user accepts an answer, the question and the answer it received are stored, and later questions can retrieve that entry. In case-based reasoning this is the retain step [Aamodt and Plaza, 1994]; in a retrieval-augmented generation (RAG) system [Lewis et al., 2020] it is a change to the retrieval half. We study only that retrieval half: the stored entry is a pointer from a past question to the thread that answered it, and nothing is generated. The idea is reasonable.

Repeat questions are common, and a stored case written in the user’s own words may match the next repeat better than the original documentation does.

The same mechanism is also a feedback loop. Whatever the system retrieved, if accepted, becomes evidence for future retrieval. Users accept wrong answers for many reasons: the answer looks plausible, they trust the top-ranked result [Joachims et al., 2005, 2007], they give up, or they solve the problem another way and close the ticket. Each accepted wrong answer adds an entry that may resemble the next question more than the correct document does. Deployed systems already write resolved tickets back into their knowledge [Liu et al., 2026, Wang et al., 2026b, Guan et al., 2026], and support practice recommends it [Consortium for Service Innovation, 2016]. Several of the underlying phenomena are known: agents that write their own outputs to memory propagate their errors [Xiong et al., 2026b], query-based document expansion adds noise that filtering can remove [Gospodinov et al., 2023], and question-to-question similarity is a strong retrieval signal [Jeon et al., 2005, Sakata et al., 2019]. What we add is a measurement of the human-gated, non-generative case on real repeat questions: where the loop turns from helpful to harmful, how that point can be predicted, and how the content of the stored entry trades harm against gain.

This paper extends a two-page pilot (two forums, one retriever, five orders) to all twelve CQADupStack forums, three retrievers, top-1 and top-5 metrics, ten random orders and 20-round streams. Version 2 added an ablation of what the stored entry contains, acceptance that depends on retrieval confidence, two baselines, an out-of-sample test of the crossover estimate, and a cross-encoder reranker. Version 3 rewrites the related work, adds a Doc2Query-- style filter and a semantic-cache baseline, and repeats the main experiment on a second corpus whose documents are not questions. The contributions are:

1. **A measurement of the sign change.** On real repeat questions, naive write-back turns from helpful to harmful at a crossover acceptance rate p^* for wrong answers. We report p^* with bootstrap intervals for 36 forum–retriever pairs and six MS MARCO configurations and show that it varies widely, so it must be measured per configuration. On CQADupStack a lexical retriever is nearly immune (Section 4); on MS MARCO it is not immune but still has the highest crossover (Section 11). The crossover moves only slightly when acceptance of wrong answers depends on retrieval confidence (Section 6).
2. **The mechanism, and the harm/gain trade-off of entry design.** The harm comes from question-to-question similarity: a stored question outscores the correct post for the next question. Storing the question together with the thread text, or reranking with a cross-encoder, removes most of the loss; the price is most of the gain (Section 5).
3. **A simple model of the crossover that can be checked.** We show that a one-parameter law based on base accuracy does not fit, and derive a plug-in estimate from an exact decomposition of the accuracy change into per-entry effects. Measured at one acceptance rate on half of the orders, it predicts the crossover on the other half (Section 9).
4. **Baselines and monitoring.** Reopening wrongly accepted answers (demotion) recovers a median of 52% of the loss where a loss remains, and does better than a confidence gate, a case-base footprint rule, a cross-encoder filter in the style of Doc2Query--, or a semantic cache with a similarity threshold (Sections 7 and 8). None of five label-free monitors separates harmful from harmless runs much better than the acceptance rate itself (Section 10).

2 Related work

We searched the literature in October 2026. We found no study of human-gated, non-generative write-back in which users accept some wrong answers, that measures the acceptance rate at which write-back turns harmful and predicts it out of sample, or that measures the trade-off between storing questions and storing questions with their threads. The broad phenomena are known, and we discuss them below.

Agents that write their own experience to memory. The closest prior work is Xiong et al. [2026b], who study LLM agents that store their own past task inputs and generated outputs and retrieve them as examples. They find *experience-following* (a new input similar to a stored one yields a similar output), show that errors in stored records propagate to later outputs, and show that selective addition (storing only records that pass a quality check) and deletion of poor records improve performance over unchecked growth. Our setting maps onto theirs: write-back is addition, our confidence gate and cross-encoder filter are forms of selective addition, and demotion, which removes entries whose answers users reopen, is a form of deletion. The differences are that our entries are not generated (they are pointers from a real question to an existing thread), that addition is gated by a simulated human who sometimes accepts wrong answers, and that we vary that acceptance rate to find where the loop changes sign. Related agent-memory work studies how stored experience is represented and transferred [Hu et al., 2026], learns which memories to reuse from environment feedback [Zhang et al., 2026a], gates or annotates admission with verifier signals [Wang et al., 2026c], repairs actions that depended on faulty memories [Yu et al., 2026a], checks whether remembered state is still valid [Xiong et al., 2026a], and benchmarks safety risks when memory evolves [Shao et al., 2026, Xie et al., 2026]. Li et al. [2026] study admission to memory shared between LLM agents and show that copied or paraphrased claims can be mistaken for independent corroboration; there the agents and admission policies are automatic, with no human acceptance. Learning from user logs must also handle noisy feedback [Wang et al., 2026a].

Self-improving RAG and write-back. Bidirectional RAG [Chinthala, 2025] writes generated answers back to the corpus only after automatic multi-stage validation (entailment, attribution, novelty); WriteBack-RAG [Lu et al., 2026] distills retrieved evidence into new knowledge units offline using labelled examples; Bang et al. [2026] add corrective feedback to RAG at inference time, treating the feedback as correct; EvoRAG [Fu et al., 2026] propagates feedback into a knowledge graph. These systems write generated or distilled content and assume validation or feedback is reliable. We study the opposite failure mode in a simpler system: nothing is generated, and the feedback (acceptance) is sometimes wrong.

Query-based document expansion and question retrieval. Storing a user question next to the document that answered it is a form of document expansion with queries. Query association [Scholer et al., 2004] attaches past queries to the documents they retrieved; doc2query and docTTTTTquery [Nogueira et al., 2019, Nogueira and Lin, 2019] append generated queries to documents; Doc2Query-- [Gospodinov et al., 2023] shows that many generated queries are noise and filters them with a relevance model before indexing, which motivates our filter baseline. Retrieval of similar questions from question-answer archives [Jeon et al., 2005] and FAQ retrieval that combines query-question similarity with query-answer relevance [Sakata et al., 2019] show that question-to-question similarity is a strong signal; our ablation measures its downside when the stored question points to a wrong answer.

Generated text feeding back into retrieval. Chen et al. [2024a] simulate rounds in which LLM-generated passages enter the corpus of open-domain QA systems and come to dominate retrieval. Dai et al. [2024] show that neural retrievers rank LLM-generated text above human-written text of the same meaning, Huang et al. [2026] show that generated text in a RAG corpus lowers retrieval quality, Yu et al. [2026b] measure retrieval collapse as generated pages contaminate a web-scale pool, Druck and Smith [2026] show that a model retrieving its own documents over-cites them, and Pan et al. [2023] study misinformation pollution. In our setting the written-back text is human-written; what dense encoders over-reward is its similarity in form (a short question) to the incoming query.

Poisoning of memories, knowledge bases and caches. PoisonedRAG [Zou et al., 2025] and AgentPoison [Chen et al., 2024b] inject crafted documents or memory records; MINJA [Dong et al., 2025] poisons an agent’s memory through queries alone; MemSentry [Roy and Basu, 2026] screens memory writes. Semantic caches [Bang, 2023] return a stored answer when a new query is similar enough to a cached one. Zhang et al. [2026b] show that a cached entry can be poisoned so that it stays similar to target queries while carrying a malicious answer, and argue that similarity is not validity; vCache [Schroeder et al., 2026] learns per-entry thresholds with an error guarantee, and other systems check evidence overlap, document versions or freshness before reuse [Shah, 2026, Zeng and Jin, 2026, Mansoor et al., 2026]. Write-back is an unintentional version of the same channel: the entries are approved by ordinary users, not crafted by an adversary. Our cache baseline (Section 8) shows the same weakness without any attacker.

Case-based reasoning. A support desk that stores resolved tickets for reuse is a case-based reasoner, and write-back is the *retain* step [Aamodt and Plaza, 1994]. Case-base maintenance deletes cases that add little competence [Smyth and Keane, 1995] and chooses which cases to add [Zhu and Yang, 1999], assuming retained cases are correct; we test a footprint-style rule as a baseline.

Feedback loops and implicit feedback. Generative models trained on their own outputs degrade [Shumailov et al., 2024, Alemohammad et al., 2024]. Sculley et al. [2015] list feedback loops as technical debt, Perdomo et al. [2020] formalise performative prediction, Chaney et al. [2018] and Jiang et al. [2019] show that recommender feedback loops homogenise or degenerate, and Ensign et al. [2018] show runaway loops in predictive policing. Clicks and acceptances are biased by rank and by trust in the system [Joachims et al., 2005, 2007], which learning to rank must correct for [Joachims et al., 2017]; simulated users with trust beliefs have been proposed for generated content [Zerhoubi, 2026]. This trust bias motivates our confidence-dependent acceptance model.

Data, retrievers and monitoring. CQADupStack [Hoogeveen et al., 2015] contains threads from twelve StackExchange forums with community duplicate links; we use the BEIR distribution [Thakur et al., 2021]. The second corpus is built from MS MARCO [Nguyen et al., 2016]. We use bge-small-en-v1.5 [Xiao et al., 2024], all-MiniLM-L6-v2 [Reimers and Gurevych, 2019, Wang et al., 2020], BM25 [Robertson and Zaragoza, 2009] and the cross-encoder ms-marco-MiniLM-L-6-v2. Shift detection without labels is studied by Rabanser et al. [2019], and disagreement between models can track error [Jiang et al., 2022]; trust propagation over documents needs some human reliability labels [Li and Chen, 2026].

Forum	KB posts	Asks	Families	Repeat asks
android	22,001	1,696	699	997
english	38,026	3,765	1,570	2,195
gaming	44,633	2,263	1,595	668
gis	37,408	1,114	885	229
mathematica	16,151	1,358	804	554
physics	37,422	1,933	1,039	894
programmers	31,377	1,675	876	799
stats	42,008	913	652	261
tex	65,936	5,154	2,906	2,248
unix	46,761	1,693	1,072	621
webmasters	16,516	1,395	506	889
wordpress	48,402	744	541	203
total	446,641	23,703	13,145	10,558

Table 1: The twelve CQADupStack forums as used here. “Repeat asks” are asks of a family beyond its first ask.

3 Setup

Problem families. For each CQADupStack test question, the community has marked one or more posts as duplicates. A question and its marked duplicates form a *problem family*. The post with the lowest identifier (the oldest) stays in the knowledge base (KB) as the canonical thread; the test question and the titles of the other marked duplicates become *asks* of the same problem. All other posts stay in the KB as distractors. An answer is correct only when the retrieved thread is the canonical thread of the ask’s own family. Every ask is a real question written by a real user, and every wrong answer is a real retrieval error. Table 1 gives the sizes.

Retrievers. KB posts are embedded as title plus the first part of the body (1,200 characters). bge-small-en-v1.5 (512 tokens) and all-MiniLM-L6-v2 (256 tokens) are run with sentence-transformers, and retrieval is cosine similarity. BM25 uses $k_1 = 1.2$, $b = 0.75$, English stop words removed. A written-back entry contains the ask text and points to the thread that was shown. For BM25, written-back entries are scored with the IDF and average length of the original KB; we do not update these statistics as entries are added, which is a small approximation.

The loop. Asks arrive in a random order split into R rounds ($R = 10$; also $R = 20$ on four forums). For each ask, the system shows the top-1 thread. A simulated user accepts a correct answer with probability $p_c = 0.9$ and a wrong one with probability $p_w \in \{0.1, 0.2, \dots, 0.9, 0.95\}$. Accepted answers are written back, and new entries become retrievable at the start of the next round (batch re-indexing). Four conditions are compared: *no write-back*; *naive write-back*; *+demotion*, where half of the accepted wrong answers are reopened in the next round and their entries removed; and *+demotion+quarantine*, where in addition any entry (original or written back) rejected twice is withdrawn. Safeguards are run at $p_w \in \{0.3, 0.6, 0.8, 0.95\}$. The acceptance decision is the only simulated part. Constant acceptance is the simplest model; real acceptance is biased toward confident, highly ranked answers [Joachims et al., 2005, 2007], which is why version 2 adds a confidence-dependent model.

Version-2 extensions. *Entry content.* Besides the ask text (v1), we test (b) a copy of the shown thread’s own text (title and body prefix, as indexed) under a new identifier, and (c) the ask text followed by the shown thread’s text, embedded or BM25-scored as one entry. Entry (b) scores exactly like the post it copies. Entries (c) are embedded for every (ask, thread) pair that occurs in any run; we iterated the simulation until no pair was missing. *Difficulty-dependent acceptance.* A wrong answer to ask i is accepted with probability $\sigma(a + 6(q_i - \frac{1}{2}))$, where q_i is the percentile of the ask’s no-write-back top-1 score within the forum and retriever, so confident wrong answers are accepted more often. The offset a is solved so that the mean probability over asks whose no-write-back answer is wrong equals the nominal p_w ; the realised acceptance rate of wrong answers at nominal 0.95 is 0.95 on average. *Baselines.* A *confidence gate* writes back an accepted answer only if its top-1 score is at least the median top-1 score of the first round (entries from round 1 are filtered once the threshold is set). A *footprint rule* in the spirit of case-base maintenance [Smyth and Keane, 1995, Zhu and Yang, 1999] keeps a new entry only if no existing entry for the same thread has ask-to-ask similarity at or above τ , the median over asks of the similarity to the nearest other ask (label-free); for entry (b) all entries for a thread are identical, so only the first is kept. Gate and footprint are run at $p_w \in \{0.3, 0.6, 0.8, 0.95\}$; demotion is run on the full grid. *Reranker.* On four forums (android, mathematica, unix, webmasters) bge retrieves the top 30 posts and the written-back entries, the 20 highest are reranked by the cross-encoder, and the top five are shown. Cross-encoder scores are cached for each ask against its top 30 posts and its top 300 other asks under bge; entries outside that set are not retrieved. *Diagnostics.* For each ask whose canonical thread is among the cached candidates we record whether some written-back entry scores above the canonical post (*capture*), and, for asks that the base retriever answers correctly, whether an entry pointing to a *wrong* thread does (*steal*); both are averaged over the second half of the stream at $p_w = 0.95$.

Version-3 extensions. *Cross-encoder filter.* In the spirit of Doc2Query-- [Gospodinov et al., 2023], each accepted (ask, shown thread) pair is scored by the cross-encoder already used in version 2 (ms-marco-MiniLM-L-6-v2, 256 tokens) before it is written back, and dropped if its score is below a threshold. The threshold is label-free: the q -quantile of the scores of the pairs accepted in round 1, $q \in \{0.25, 0.5, 0.75\}$; it is fixed after round 1 and the round-1 entries are filtered with it. The filter is run for question entries and question-plus-thread entries on all 36 forum–retriever pairs over the full p_w grid. Cross-encoder scores were computed for every pair that occurs in any run (iterated until none was missing). *Semantic cache.* Stored question entries are used as a cache: for an incoming ask, if its highest similarity (the deployed retriever’s score) to any stored question is at least τ , the stored thread is shown; otherwise the base retriever answers from the original posts only. As in a semantic cache, only accepted answers to cache misses are stored. τ is the q -quantile, $q \in \{0.5, 0.75, 0.9, 0.95\}$, of each ask’s similarity to its nearest other ask (label-free). *Second corpus.* See Section 11. All version-3 runs use the same ten orders and random streams as version 2; the version-3 code reproduces the version-2 naive runs exactly (780 runs compared, 0 differences).

Metrics and statistics. We report top-1 accuracy over the whole stream and hit@5 (whether any of the top five entries points to the correct thread; the loop itself is driven by top-1). Each configuration is run with ten random orders. We report the paired difference from no write-back on the same order, in percentage points, with a 95% t -interval over the ten orders. The crossover p^* is the first p_w at which the mean paired difference becomes negative, by linear interpolation on the grid; its 95% interval comes from 1,000 bootstrap resamples of the orders. A crossover below the grid is reported as <0.10 , and one above it as >0.95 .

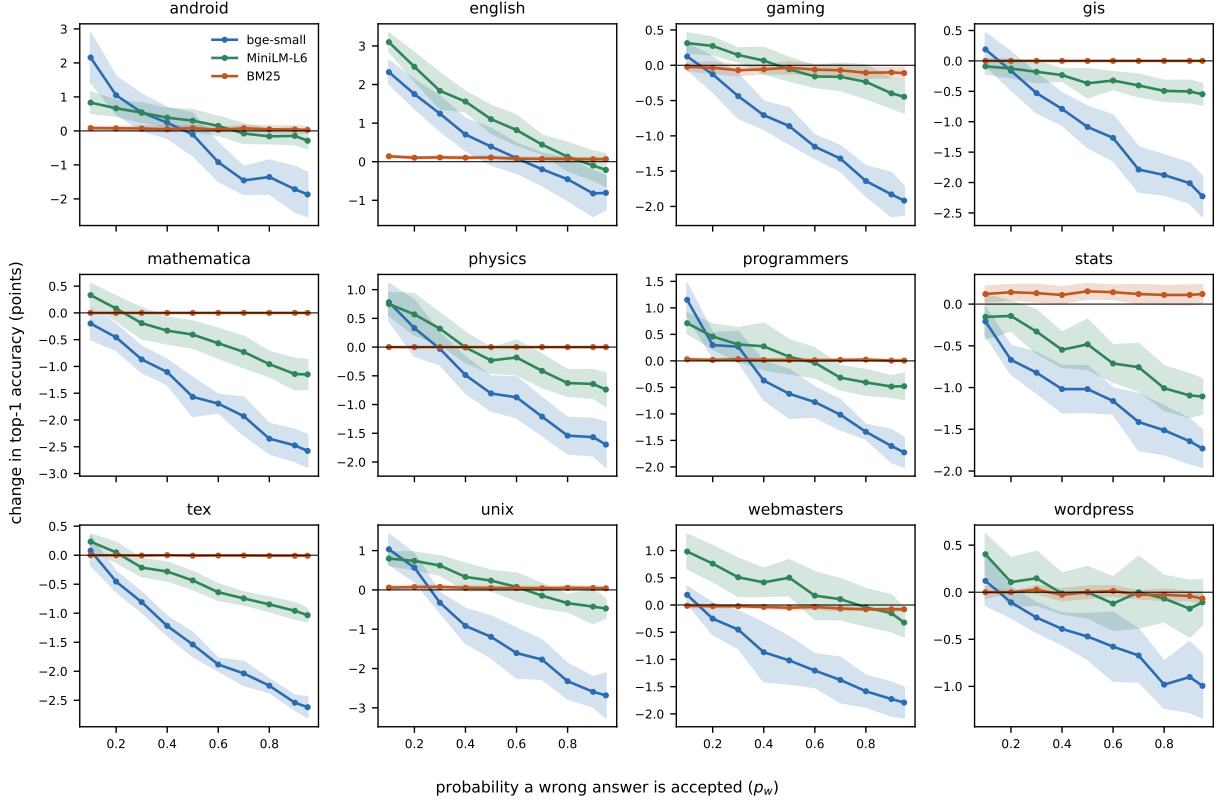


Figure 1: Change in top-1 accuracy over the stream from naive write-back, relative to no write-back on the same order, as the acceptance rate of wrong answers grows. Bands are 95% intervals over ten orders. Ten rounds.

4 Results

The loop changes sign. Figure 1 shows the paired change in top-1 accuracy for every forum and retriever. For both dense retrievers, the curve falls almost linearly with p_w (median R^2 of a linear fit above 0.95). When users reject most wrong answers, write-back helps on most forums, by up to 3.1 points. When they accept nearly all of them, it hurts: at $p_w = 0.95$ the change for dense retrievers ranges from -2.68 to -0.11 points, and the 95% interval lies entirely below zero in 22 of 24 dense configurations. With base top-1 accuracies between 9% and 35% (Table 2), a loss of 2.7 points is roughly a tenth of all correct answers. The pilot’s numbers for Unix and Android are reproduced in direction and size (the pilot reported -2.1 and -1.8 points at $p_w = 0.95$ with five orders; here -2.68 ± 0.57 and -1.87 ± 0.65 with ten orders, a re-computed embedding, and batch semantics that match the pilot).

The crossover is not a constant. Table 2 lists p^* . Among dense configurations with an interior crossover it ranges from 0.11 to 0.86. On some forums (stats, mathematica with bge, gis with MiniLM) write-back is harmful at every tested acceptance rate. The weaker MiniLM model often has a *higher* crossover than bge on the same forum, so a better base retriever does not by itself make write-back safer. Crossovers for hit@5 are broadly similar but noisier. The intervals are wide in places (for example MiniLM on wordpress), because the curve is nearly flat there.

Forum	Retriever	Top-1	Hit@5	Observed p^* (top-1)	Observed p^* (hit@5)	Plug-in $\hat{p}^*$
android	bge	16.7	32.8	0.47 [0.37, 0.54]	0.55 [0.39, 0.63]	0.39
	MiniLM	22.3	38.9	0.67 [0.56, 0.78]	0.67 [0.48, 0.89]	0.69
	BM25	13.7	23.0	>0.95	>0.95	>0.95
english	bge	19.1	34.2	0.63 [0.50, 0.75]	0.39 [0.33, 0.44]	0.61
	MiniLM	23.0	40.6	0.86 [0.77, 0.95]	0.53 [0.46, 0.60]	0.85
	BM25	11.2	17.9	>0.95	0.59 [0.37, 0.95]	>0.95
gaming	bge	31.7	54.3	0.15 [0.10, 0.22]	<0.10	0.14
	MiniLM	34.7	57.0	0.45 [0.35, 0.56]	<0.10	0.49
	BM25	27.0	41.1	<0.10	<0.10	<0.10
gis	bge	21.4	39.0	0.15 [0.10, 0.22]	0.19 [0.16, 0.34]	0.14
	MiniLM	25.2	44.4	<0.10	0.31 [0.22, 0.41]	<0.10
	BM25	16.8	29.9	>0.95	0.10 [0.10, 0.40]	–
mathematica	bge	12.1	26.1	<0.10	0.12 [0.10, 0.19]	<0.10
	MiniLM	13.0	28.0	0.23 [0.17, 0.31]	0.20 [0.15, 0.31]	0.24
	BM25	9.4	18.7	>0.95	<0.10	–
physics	bge	20.2	36.1	0.29 [0.19, 0.34]	0.25 [0.18, 0.29]	0.29
	MiniLM	21.5	40.4	0.40 [0.33, 0.47]	0.28 [0.16, 0.37]	0.45
	BM25	13.5	24.3	>0.95	0.20 [0.10, 0.40]	–
programmers	bge	16.4	33.3	0.34 [0.30, 0.39]	0.20 [0.17, 0.30]	0.36
	MiniLM	17.4	34.3	0.57 [0.33, 0.66]	0.56 [0.36, 0.73]	0.47
	BM25	11.6	20.1	>0.95	<0.10	>0.95
stats	bge	18.1	31.5	<0.10	0.16 [0.11, 0.19]	<0.10
	MiniLM	20.2	35.7	<0.10	0.35 [0.31, 0.41]	0.11
	BM25	15.8	24.9	>0.95	>0.95	–
tex	bge	11.5	22.7	0.11 [0.10, 0.14]	0.14 [0.11, 0.16]	0.13
	MiniLM	14.4	28.4	0.22 [0.16, 0.27]	0.26 [0.21, 0.32]	0.17
	BM25	10.6	18.7	0.10 [0.10, 0.40]	<0.10	<0.10
unix	bge	20.8	36.3	0.26 [0.24, 0.29]	0.28 [0.26, 0.31]	0.24
	MiniLM	23.1	44.1	0.63 [0.50, 0.74]	0.55 [0.45, 0.65]	0.73
	BM25	14.1	22.3	>0.95	>0.95	>0.95
webmasters	bge	10.8	21.1	0.14 [0.11, 0.19]	0.16 [0.13, 0.19]	0.16
	MiniLM	12.3	24.9	0.77 [0.56, 0.93]	0.59 [0.50, 0.78]	0.69
	BM25	9.3	16.1	<0.10	0.85 [0.57, 0.95]	<0.10
wordpress	bge	14.0	27.8	0.15 [0.10, 0.20]	<0.10	0.13
	MiniLM	16.9	31.5	0.39 [0.17, 0.95]	>0.95	0.48
	BM25	14.8	24.1	0.35 [0.10, 0.87]	0.10 [0.10, 0.50]	0.61

Table 2: No-write-back accuracy (%) and the crossover acceptance rate p^* for naive write-back, with 95% bootstrap intervals over ten orders. The last column is the plug-in prediction of Section 9, calibrated from the runs at $p_w = 0.3$ only. “–”: no entry ever changed an answer.

BM25 is nearly immune here. With BM25, write-back changes accuracy by less than 0.2 points in every forum and condition, and written-back entries supply at most 8% of top-1 answers in the second half of the stream, against 15–72% for dense retrievers at $p_w = 0.95$. The reason is visible before any simulation. For each ask we compare its score against the best original post with its score against the best other ask from the same family, as if that ask had been written back. Under bge, the other ask scores higher for 36–72% of asks (depending on forum); under MiniLM for 20–49%; under BM25 for only 2–12%. Short titles share few terms with a new title and lose to long posts that match more terms. Dense encoders place two short titles close together whether or not they describe the same problem: under bge an ask from a *different* family outscores the best original post for 33–74% of asks. This is the mechanism of the takeover, and it also explains why most of the harm comes from wrong entries capturing questions they do not belong to. We do not claim BM25 is safe in general: on the MS MARCO corpus (Section 11) BM25 entries do take over part of the answers, and served as a cache (Section 8) BM25 entries are harmful.

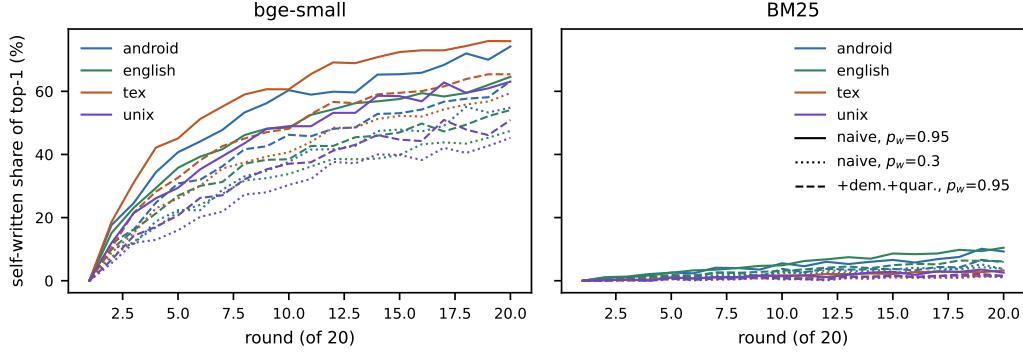


Figure 2: Share of top-1 answers that come from written-back entries over 20 rounds, averaged over ten orders. Left: bge-small. Right: BM25.

Forum	Retriever	p^* (10 rounds)	p^* (20 rounds)	Naive, $p_w=0.95$ (10)	(20)
android	bge	0.47 [0.37, 0.54]	0.45 [0.36, 0.53]	-1.87 ± 0.65	-2.04 ± 0.69
	MiniLM	0.67 [0.56, 0.78]	0.66 [0.55, 0.91]	-0.29 ± 0.24	-0.29 ± 0.25
english	bge	0.63 [0.50, 0.75]	0.61 [0.50, 0.70]	-0.81 ± 0.44	-0.84 ± 0.42
	MiniLM	0.86 [0.77, 0.95]	0.82 [0.77, 0.95]	-0.21 ± 0.42	-0.22 ± 0.48
tex	bge	0.11 [0.10, 0.14]	0.12 [0.10, 0.15]	-2.62 ± 0.18	-2.73 ± 0.22
	MiniLM	0.22 [0.16, 0.27]	0.22 [0.17, 0.27]	-1.03 ± 0.11	-1.08 ± 0.10
unix	bge	0.26 [0.24, 0.29]	0.26 [0.22, 0.28]	-2.68 ± 0.57	-2.95 ± 0.57
	MiniLM	0.63 [0.50, 0.74]	0.63 [0.51, 0.77]	-0.47 ± 0.24	-0.47 ± 0.32

Table 3: Crossover and loss at $p_w = 0.95$ with 10 and 20 rounds on four forums.

Self-written entries take over retrieval. Figure 2 shows the takeover over 20 rounds. With bge and naive write-back at $p_w = 0.95$, written-back entries supply more than half of all top-1 answers by round 9 to 13 (of 20) on all four forums, and 61–76% by the end. Safeguards slow the takeover but do not stop it.

Longer streams. Doubling the number of rounds to 20 on four forums leaves crossovers and losses within the intervals of the 10-round runs (Table 3). The stream length in asks is the same; only the re-indexing interval is shorter. A longer stream of new asks would need more data than CQADupStack provides.

Safeguards recover part of the loss. Table 11 (Appendix A) reports the paired change at $p_w = 0.3$ and 0.95 for the dense retrievers. At $p_w = 0.3$, write-back with demotion and quarantine is positive in 20 of 24 dense configurations. At $p_w = 0.95$ it turns the loss into a non-negative change in 9 of 24 configurations. In the other 15 it recovers a median of 52% of the loss (range 32–76%), so the loss is reduced but not removed. Quarantine adds very little on top of demotion: the two differ by at most 0.10 points at $p_w = 0.95$. With a 90% acceptance rate for correct answers, few entries are rejected twice before they have done their damage.

5 What the stored entry contains

Version 1 stored the ask text. Table 4 and Figure 3 compare the three entry contents; per-configuration crossovers are in Table 12 (Appendix).

Retriever	Entry content	Δ @0.3	Δ @0.95	sig. < 0 @0.95	median p^*	self-share @0.95	capture	steal
bge	ask text (v1)	-0.21	-1.89	12/12	0.15	56%	74%	26.3%
	shown thread text	+0.00	+0.00	0/12	>0.95	0%	51%	0.0%
	ask + thread text	+0.14	-0.02	1/12	0.75	14%	52%	2.7%
MiniLM	ask text (v1)	+0.29	-0.58	10/12	0.43	33%	58%	12.2%
	shown thread text	+0.00	+0.00	0/12	>0.95	0%	47%	0.0%
	ask + thread text	+0.20	-0.02	4/12	0.80	18%	53%	3.5%
BM25	ask text (v1)	+0.03	-0.00	4/12	>0.95	4%	30%	0.6%
	shown thread text	+0.00	+0.00	0/12	>0.95	0%	47%	0.0%
	ask + thread text	+0.07	+0.01	2/12	>0.95	13%	52%	1.6%

Table 4: Entry-content ablation, naive write-back. Δ : mean over the twelve forums of the paired change in top-1 accuracy (points) at $p_w = 0.3$ and 0.95. “sig. < 0”: forums whose 95% interval at $p_w = 0.95$ is below zero. Median p^* over forums (crossovers below or above the grid counted at the grid ends). Self-share: share of top-1 answers from written-back entries, second half of the stream. Capture and steal: see Section 3.

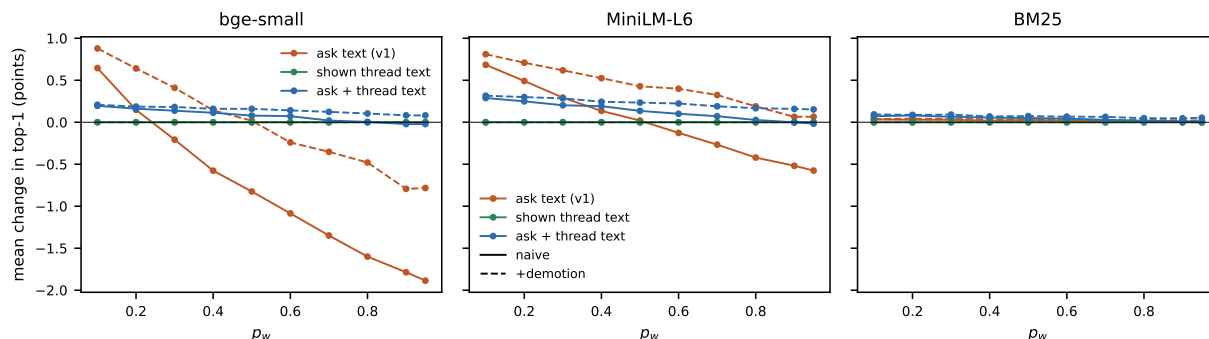


Figure 3: Mean paired change in top-1 accuracy over the twelve forums for the three entry contents, naive (solid) and with demotion (dashed).

Question entries steal answers. With question entries and bge, a wrong entry scores above the canonical post on 26% of the asks that the base retriever answers correctly (MiniLM 12%, BM25 1%). That is the harm channel: the incoming question is a short title, and another short title, even about a different problem, is closer to it than any full post. The ordering of retrievers by steal rate matches their ordering by harm.

Thread copies are inert for top-1. A copy of the shown thread scores exactly like the original post, so it can never rank above it, and the top-1 thread never changes: the change in top-1 accuracy is zero by construction at every p_w . Capture is still about half, but only because copies of a wrong post that already outranked the canonical one inherit its score; steal is zero. Copies do crowd the top five: hit@5 falls by 1.2 points on average at $p_w = 0.95$ for both dense retrievers. This variant gives neither the gain nor the loss of write-back.

Question plus thread text keeps a little of the gain and little of the loss. Concatenating the ask with the thread text puts the entry between a question and a post. Steal falls to 3% (bge) and 3% (MiniLM). The mean change at $p_w = 0.95$ moves from -1.89 to -0.02 points for bge and from -0.58 to -0.02 for MiniLM; the loss is significant in 1 (bge) and 4 (MiniLM) of twelve forums instead of 12 and 10. The median crossover rises from 0.15 to 0.75 (bge) and from 0.43 to 0.80 (MiniLM). The gain at low p_w also shrinks: at $p_w = 0.1$ the mean change falls from 0.64 to 0.20 points (bge) and from 0.68 to 0.29 (MiniLM), and the largest single-forum gain from 2.3 to 0.6

Forum	Retriever	Top-1	Hit@5	p^*	$\Delta@0.95$	self-share @0.95
android	bge	16.7	32.8	0.47 [0.37, 0.54]	-1.87 ± 0.65	65%
	bge+CE	17.6	31.4	>0.95	$+0.24 \pm 0.12$	10%
mathematica	bge	12.1	26.1	<0.10	-2.58 ± 0.30	65%
	bge+CE	12.3	25.5	0.73 [0.29, 0.93]	-0.13 ± 0.11	7%
unix	bge	20.8	36.3	0.26 [0.24, 0.29]	-2.68 ± 0.57	56%
	bge+CE	20.0	34.0	>0.95	$+0.18 \pm 0.14$	6%
webmasters	bge	10.8	21.1	0.14 [0.12, 0.19]	-1.79 ± 0.28	72%
	bge+CE	11.7	20.4	0.81 [0.19, 0.95]	-0.02 ± 0.15	19%

Table 5: bge alone and bge with cross-encoder reranking (top 20), question entries, naive write-back. Self-share in the second half of the stream.

(bge) and 3.1 to 1.0 points (MiniLM). The curve is flatter rather than shifted: at $p_w = 0.3$ the concatenated entries do better than question entries for bge (+0.14 against -0.21) and slightly worse for MiniLM (+0.20 against +0.29). With demotion the concatenated entries are non-negative on average across the grid. BM25 stays essentially unaffected with all three entry types, including the longer concatenated entries that v1 flagged as untested.

The trade-off is the point. The same property that makes a stored question useful, that it looks like the next question, is what lets a wrong one capture answers. Entry designs that reduce one reduce the other. In this data no design gave the full gain of question entries without their loss.

A cross-encoder reranker. We tried to raise base accuracy with a cross-encoder over the bge candidates (Table 5). It did not: top-1 accuracy changed by -0.8 to 0.9 points (the reranker is trained on web queries and passages, not forum titles). It did change the loop. Reranked, written-back questions supply 6–19% of top-1 answers at $p_w = 0.95$ instead of 56–72%, and the change at $p_w = 0.95$ lies between -0.13 and $+0.24$ points (significantly negative on 1 of 4 forums, mathematica), against -1.8 to -2.7 for bge alone. The steal rate falls from 30–39% under bge to at most 1.1% after reranking: the cross-encoder does not give a wrong stored title the advantage over the correct post that the bi-encoder does. Because base accuracy did not rise, this run does not tell us whether the effect holds at higher base accuracy; that question remains open.

6 Acceptance that depends on confidence

If users accept wrong answers more readily when the system is confident, the accepted wrong answers are concentrated on asks where the retriever is confidently wrong, and these are the asks whose entries are most likely to resemble future asks. Table 13 (Appendix) and Figure 4 compare crossovers under constant and confidence-dependent acceptance with the same average rate. For question entries and dense retrievers, the crossover moves down in most configurations (median shift -0.06, range -0.27 to +0.19); each crossover lies outside the other’s bootstrap interval, or censoring changes, downwards in 7 of 24 configurations and upwards in 0. The mean loss at $p_w = 0.95$ is nearly unchanged (-1.23 versus -1.27 points) because at that rate almost all wrong answers are accepted either way. The qualitative picture of Section 4 holds; crossovers measured under constant acceptance are, if anything, slightly optimistic.

7 Safeguards and baselines

Table 6 compares the v1 safeguards with the two new baselines. For question entries, demotion is the strongest: at $p_w = 0.95$ it reduces the mean loss from -1.23 to -0.36 points and is non-negative

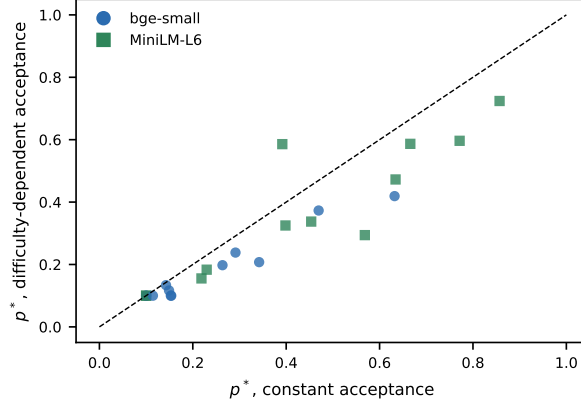


Figure 4: Crossover under constant against confidence-dependent acceptance of wrong answers, question entries, dense retrievers. Points at 0.05 and 1.0 are below or above the grid.

Entry content	Rule	$\Delta@0.3$	$\Delta@0.6$	$\Delta@0.8$	$\Delta@0.95$	$\geq 0 @0.95$	entries kept
ask text (v1)	naive	+0.04	-0.61	-1.01	-1.23	0/24	100%
	+demotion	+0.51	+0.08	-0.15	-0.36	9/24	100%
	+dem.+quar.	+0.65	+0.17	-0.13	-0.37	9/24	100%
	confidence gate	+0.16	-0.34	-0.67	-0.83	2/24	63%
	footprint rule	-0.04	-0.59	-0.93	-1.09	0/24	79%
shown thread text	naive	+0.00	+0.00	+0.00	+0.00	24/24	100%
	+demotion	+0.00	+0.00	+0.00	+0.00	24/24	100%
	+dem.+quar.	+0.05	+0.02	+0.00	-0.00	16/24	100%
	confidence gate	+0.00	+0.00	+0.00	+0.00	24/24	55%
	footprint rule	+0.00	+0.00	+0.00	+0.00	24/24	85%
ask + thread text	naive	+0.17	+0.09	+0.02	-0.02	7/24	100%
	+demotion	+0.23	+0.18	+0.14	+0.12	20/24	100%
	+dem.+quar.	+0.29	+0.21	+0.14	+0.11	20/24	100%
	confidence gate	+0.14	+0.08	+0.04	+0.02	14/24	56%
	footprint rule	+0.13	+0.05	-0.01	-0.04	8/24	90%

Table 6: Mean paired change in top-1 accuracy over the 24 dense configurations for each entry content and safeguard. “ ≥ 0 ”: configurations with a non-negative mean change at $p_w = 0.95$. Entries kept: written-back entries at the end of the stream relative to naive, at $p_w = 0.95$.

in 9 of 24 configurations; quarantine adds little (-0.37). The confidence gate keeps 63% of the entries and reduces the loss to -0.83 points, with smaller gains at low p_w (+0.16 at 0.3, against +0.51 for demotion). The footprint rule keeps 79% of the entries and barely changes the outcome (-1.09 at 0.95, -0.04 at 0.3): it removes entries that resemble existing entries for the same thread, whether those entries are right or wrong, so it does not target the entries that cause harm. Both rules use only signals available without labels, but neither targets wrong entries; demotion does, through the user who reopens. With question-plus-thread entries every rule is close to zero, and demotion again does best (+0.12 at 0.95, non-negative in 20 of 24).

8 A relevance filter and a semantic cache

Cross-encoder filter. Table 7 and Figure 5 compare the filter with the rules of Section 7. For question entries the filter does what Doc2Query-- does for generated queries: it removes wrong entries more often than correct ones. At $q = 0.5$ it keeps 79% of correct entries and 38% of wrong

Entry content	Rule	$\Delta@0.3$	$\Delta@0.95$	$\geq 0@0.95$	sig. $< 0@0.95$	median p^*	kept	correct kept	wrong kept
ask text	naive	+0.04	-1.23	0/24	22/24	0.28	100%	100%	100%
	+demotion	+0.51	-0.36	9/24	14/24	0.60	100%	105%	99%
	confidence gate	+0.16	-0.83	2/24	18/24	—	63%	78%	60%
	footprint rule	-0.04	-1.09	0/24	22/24	—	79%	89%	77%
	CE filter, $q = 0.25$	+0.15	-0.87	1/24	19/24	0.34	67%	92%	62%
	CE filter, $q = 0.5$	+0.20	-0.59	4/24	15/24	0.39	45%	79%	38%
	CE filter, $q = 0.75$	+0.14	-0.30	6/24	15/24	0.28	24%	57%	17%
ask + thread text	naive	+0.17	-0.02	7/24	5/24	0.75	100%	100%	100%
	+demotion	+0.23	+0.12	20/24	1/24	>0.95	100%	100%	100%
	confidence gate	+0.14	+0.02	14/24	4/24	—	56%	73%	52%
	footprint rule	+0.13	-0.04	8/24	6/24	—	90%	91%	89%
	CE filter, $q = 0.25$	+0.15	+0.02	14/24	3/24	>0.95	74%	91%	70%
	CE filter, $q = 0.5$	+0.11	+0.03	15/24	2/24	>0.95	50%	77%	44%
	CE filter, $q = 0.75$	+0.05	+0.02	17/24	2/24	>0.95	25%	55%	19%

Table 7: Write-back rules over the 24 dense configurations (CQADupStack). Δ : mean paired change in top-1 accuracy (points). “ ≥ 0 ”: configurations with a non-negative mean change at $p_w = 0.95$; “sig. < 0 ”: 95% interval below zero. Median p^* over configurations (crossovers outside the grid counted at the grid ends); gate and footprint were run only at $p_w \in \{0.3, 0.6, 0.8, 0.95\}$, so they have no p^* . Kept: written-back entries at the end of the stream relative to naive at $p_w = 0.95$; correct and wrong kept: entries written that point to the correct or a wrong thread, relative to naive (demotion removes entries after they are written, so its columns are near 100%; for the gate, round-1 entries dropped after the threshold is set are counted as kept, as in version 2).

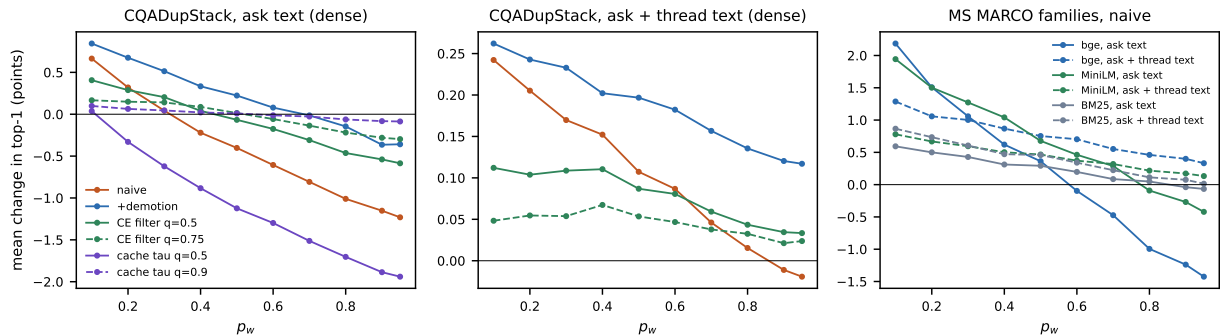


Figure 5: Mean paired change in top-1 accuracy against p_w . Left and middle: CQADupStack, mean over the 24 dense configurations, for naive write-back, demotion, the cross-encoder filter and (question entries only) the semantic cache. Right: MS MARCO families, naive write-back, per retriever.

ones, reduces the mean loss at $p_w = 0.95$ from -1.23 to -0.59 points (bge -1.89 to -0.96, MiniLM -0.58 to -0.21), and raises the gain at $p_w = 0.3$ from +0.04 to +0.20. A stricter threshold ($q = 0.75$) reduces the loss further (-0.30) but keeps only 24% of entries and less of the gain. The loss remains significant in 15 ($q = 0.5$) and 15 ($q = 0.75$) of 24 configurations, and the median crossover moves little (0.28 naive, 0.34, 0.39 and 0.28 for the three thresholds), because the filter removes part of the gain together with the loss. Demotion is better at both ends (+0.51 at 0.3, -0.36 at 0.95, non-negative in 9 configurations against 6 for the strictest filter): it acts on the user’s later rejection, which is more informative about a specific wrong answer than a relevance score trained on web queries. The filter is better than the confidence gate (-0.83) and the footprint rule (-1.09). For question-plus-thread entries every rule is within a few hundredths of a point of zero at $p_w = 0.95$. With BM25 every rule changes accuracy by at most 0.22 points in absolute value at $p_w = 0.95$.

Semantic cache. Table 8 reports the cache baseline. Its hit precision is below the base retriever’s top-1 accuracy at every threshold we tried (for bge, 6% at $q = 0.5$ and 13% at $q = 0.95$, against

Retriever	τ (quantile)	τ	hit rate	hit prec.	$\Delta@0.3$	$\Delta@0.95$	sig. < 0@0.95	median p^*
bge	naive write-back	—	—	—	-0.21	-1.89	12/12	0.15
	0.5	0.81	25.7%	6%	-0.59	-1.85	12/12	0.13
	0.75	0.85	11.9%	9%	-0.02	-0.53	12/12	0.27
	0.9	0.89	4.5%	12%	+0.05	-0.08	4/12	0.50
	0.95	0.91	2.2%	13%	+0.03	-0.01	1/12	0.78
MiniLM	naive write-back	—	—	—	+0.29	-0.58	10/12	0.43
	0.5	0.62	26.0%	9%	-0.65	-2.03	12/12	0.11
	0.75	0.71	12.0%	13%	-0.03	-0.61	12/12	0.28
	0.9	0.80	4.6%	16%	+0.04	-0.09	7/12	0.51
	0.95	0.85	2.3%	19%	+0.03	-0.01	3/12	0.56
BM25	naive write-back	—	—	—	+0.03	-0.00	4/12	>0.95
	0.5	10.74	26.4%	3%	-1.31	-2.62	12/12	0.10
	0.75	13.56	12.2%	5%	-0.42	-1.07	12/12	0.10
	0.9	16.80	4.8%	8%	-0.04	-0.31	12/12	0.14
	0.95	19.40	2.3%	9%	-0.02	-0.13	10/12	0.12

Table 8: Stored questions served as a semantic cache, CQADupStack, mean over twelve forums. τ : similarity threshold (the q -quantile of each ask’s similarity to its nearest other ask; BM25 scores are unnormalised). Hit rate: share of asks answered from the cache and hit precision: share of cache hits that are correct, both at $p_w = 0.95$. Δ : paired change in top-1 accuracy (points). The naive rows repeat ordinary write-back with question entries for comparison. Base top-1 accuracy: bge 17.7%, MiniLM 20.3%, BM25 14.0%.

17.7% base accuracy), so on average a cache hit is less likely to be right than the answer it replaces. At loose thresholds the cache is harmful even when users reject most wrong answers: at $q = 0.5$ the change at $p_w = 0.3$ is -0.59 (bge), -0.65 (MiniLM) and -1.31 (BM25), and at $p_w = 0.95$ -1.85, -2.03 and -2.62. At strict thresholds ($q = 0.95$) the cache answers 2.2% of asks and changes accuracy by a few hundredths of a point either way. We found no threshold at which the cache gave a gain comparable to ordinary write-back at low p_w . BM25 is instructive. Under ordinary write-back a stored BM25 question must outscore the original posts and rarely does (Section 4); served as a cache it bypasses that competition, and BM25 becomes as vulnerable as the dense retrievers (median p^* 0.10 to 0.12). This is the point made by Zhang et al. [2026b] for adversarial cache poisoning, that similarity to a stored query is not evidence that the stored answer is valid, observed here with ordinary users and no attacker. Per-entry thresholds [Schroeder et al., 2026] or checks of the stored answer against fresh evidence [Shah, 2026] address exactly this gap; we did not test them.

9 A model of the crossover

An exact decomposition. Under naive write-back no original post is ever withdrawn, so an answer can differ from the no-write-back answer only when the top-1 result is a written-back entry. Attributing each such change to the entry that caused it gives an exact identity

$$N \Delta = C \bar{e}_c + W \bar{e}_w, \quad (1)$$

where N is the number of asks, Δ the change in accuracy, C and W the numbers of correct and wrong entries that were written and became retrievable, and $\bar{e}_c$ and $\bar{e}_w$ the average net number of answers each kind of entry turned from wrong to right (minus right to wrong). We count these quantities directly in the simulation.

First-order approximation. If the system’s accuracy on the asks that generate entries stays near its no-write-back accuracy A , then $C \approx p_c A N'$ and $W \approx p_w (1 - A) N'$, with N' the number of

asks whose entries become retrievable. If the per-entry effects do not depend on p_w , setting $\Delta = 0$ gives

$$p^* \approx \frac{p_c A}{1 - A} \cdot \frac{\bar{e}_c}{-\bar{e}_w}. \quad (2)$$

Both assumptions are approximations: entries compete with each other, so per-entry effects shrink as more entries are written, and write-back itself changes the accuracy of later asks.

A one-parameter law does not fit. If the ratio $\kappa = \bar{e}_c/(-\bar{e}_w)$ were the same everywhere, p^* would be a function of base accuracy alone. Fitting a single κ by least squares to the 22 interior crossovers gives $\kappa = 1.60$ but $R^2 = -0.19$ (Spearman $\rho = 0.48$; Figure 6a). Base accuracy alone does not predict when write-back becomes harmful. Gaming has the highest base accuracy and one of the lowest crossovers for bge.

A plug-in estimate does. The ratio κ is specific to the forum and retriever: in our runs $\bar{e}_c$ ranges from about zero to 0.15 answers gained per correct entry, while $\bar{e}_w$ lies between -0.01 and -0.08 (dense retrievers, $p_w = 0.3$). Measuring $\bar{e}_c$ and $\bar{e}_w$ from the runs at a single acceptance rate ($p_w = 0.3$) and inserting them into Eq. (2) predicts the crossover found from the full sweep with mean absolute error 0.049 and Spearman $\rho = 0.95$ over the 21 configurations where both are interior. Prediction and observation agree, within the bootstrap interval (± 0.05) or in the direction of censoring, for 30 of 32 configurations where the estimate is defined (Figure 6b, Table 2). Calibrating at $p_w = 0.5$ or 0.95 instead gives mean absolute errors of 0.038 and 0.059.

Out of sample. The calibration above uses the same orders as the sweep it predicts. We therefore split the ten orders into two halves, estimate $\bar{e}_c$ and $\bar{e}_w$ at $p_w = 0.3$ on one half, and compare the prediction with the crossover observed on the other half, in both directions (all 36 configurations plus the four reranker configurations). Over the 39 cases where both are interior, the mean absolute error is 0.085 and Spearman $\rho = 0.88$; prediction and observation fall in the same class (below, inside or above the grid) in 54 of 64 cases where the estimate is defined. Calibrating at $p_w = 0.5$ gives 0.102 ($\rho = 0.79$). The out-of-sample error is larger than the in-sample 0.049 of version 1 but the ranking is preserved. For question-plus-thread entries the estimate fails (MAE 0.248, $\rho = 0.00$): the per-entry effects are close to zero and too noisy to estimate from five orders, and the crossover itself is poorly determined because the curve is nearly flat.

This is an empirical check of a first-order model, not a proof. It does have a practical reading. The per-entry effects can be estimated in deployment from a small audited sample of written-back entries, and the deployment’s own acceptance rate for wrong answers can then be compared against the predicted p^* .

10 Monitoring without labels

In deployment the true error is unknown. The pilot tested one label-free signal and it failed. Here we test five, all computed without ground truth: (i) the *acceptance rate*, which any deployment observes and which serves as a baseline; (ii) the *self-written share* of top-1 answers; (iii) the pilot’s signal, the share of answers where a written-back entry wins and the best original post points to a different thread; (iv) *disagreement with an independent retriever*, the share of asks whose deployed top-1 thread differs from that of a static retriever of a different kind (BM25 for dense deployments, bge for BM25), together with its rise since round 1; (v) *probe-bank drift*, the share of a fixed bank of 200 asks, queried read-only after each round, whose top-1 thread has changed since the start.

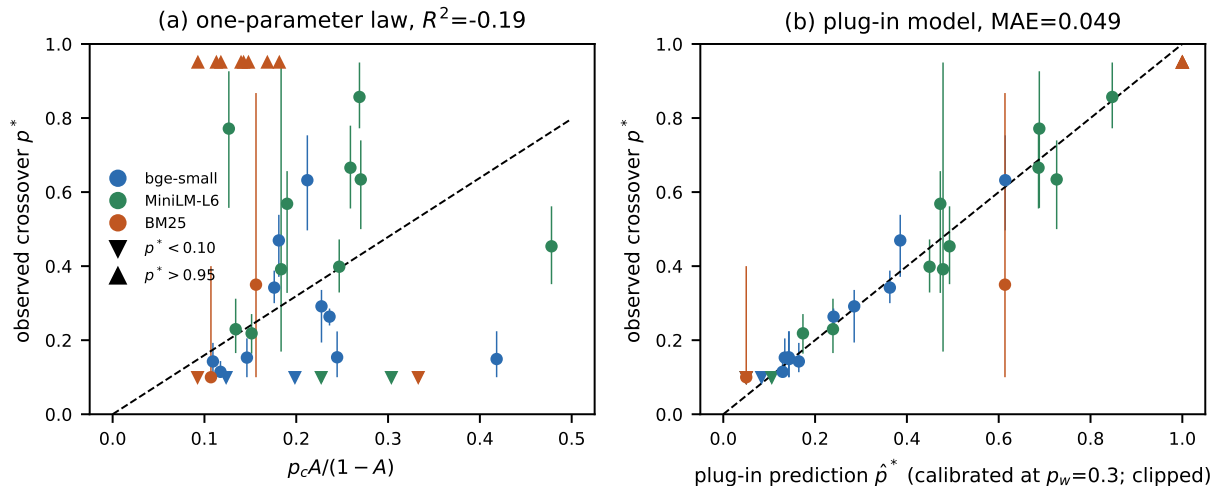


Figure 6: Observed crossover against (a) the one-parameter law $p^* = \kappa p_c A / (1 - A)$ and (b) the plug-in estimate of Eq. (2) calibrated at $p_w = 0.3$. Triangles mark crossovers outside the tested grid; bars are 95% bootstrap intervals.

Signal	ρ (all runs)	ρ (fixed p_w)	AUROC r2	AUROC r5	AUROC r10
Acceptance rate (baseline)	-0.41	+0.06	0.70	0.70	0.70
Self-written share of top-1	-0.51	-0.34	0.69	0.73	0.73
Pilot signal: self vs. human top entry	-0.51	-0.34	0.67	0.74	0.73
Disagreement with independent retriever	-0.32	-0.21	0.54	0.58	0.60
rise since round 1	-0.13	-0.07	0.54	0.57	0.58
Probe-bank drift	-0.44	-0.29	0.71	0.76	0.77

Table 9: Label-free signals against the true change in accuracy. ρ : Spearman correlation with the accuracy change (negative means the signal rises as accuracy falls), averaged over forum–retriever pairs. AUROC for identifying harmful runs from the signal after round 2, 5 and 10. Ten rounds.

For each forum and retriever, a run (one condition, acceptance rate and order) is labelled harmful if its accuracy over the second half of the stream is below no write-back on the same order. We report the Spearman correlation between each signal (averaged over the second half) and the true accuracy change, pooled over all runs and within a fixed p_w , and the AUROC for harmful runs using the signal value at a given round. All values are averaged over the 36 forum–retriever pairs (Table 9, Figure 7).

Results. The acceptance rate alone gives an AUROC of about 0.70 from the first round, because harm is driven mostly by how often wrong answers are accepted. The self-written share, the pilot signal and probe-bank drift reach 0.72–0.77 after three to five rounds, a small gain over the baseline that arrives late. Within a fixed acceptance rate their correlation with the true accuracy change is weak ($|\rho| \leq 0.34$). Disagreement with an independent retriever is close to chance (AUROC 0.50–0.61), and its rise since round 1 even more so. The reason is that dense and lexical retrievers already disagree on most asks before any write-back: under bge and BM25 the top-1 thread differs for about 80% of Android asks. Changes caused by the loop are small against that background. In summary, none of these monitors detects harm in a way that adds much to the acceptance rate. The independent-retriever monitor, which we expected to work best because it does not share the

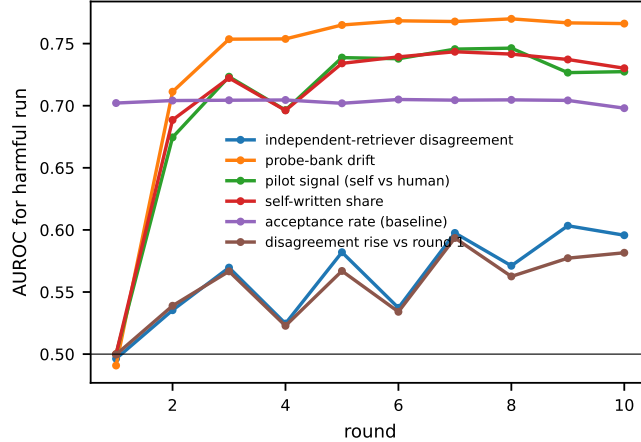


Figure 7: AUROC for identifying harmful runs from each label-free signal, by round.

deployed retriever’s errors, did not.

11 A second corpus: MS MARCO query families

CQADupStack documents are themselves questions, which may inflate question-to-question effects. We therefore looked for public data with distinct real user queries that share a relevant document that is not a question. FiQA-2018 has no passage judged relevant to two training queries; NFCorpus and TREC-COVID have many relevant documents per query, so “families” would not be well defined; SciFact claims are written by annotators. In the MS MARCO passage-ranking training qrels [Nguyen et al., 2016], however, some passages are judged relevant to two or more distinct Bing queries.

Construction. A family is a passage judged relevant to at least two queries, restricted to queries whose only judged-relevant passage is that passage, with exact duplicate query strings removed. We sampled 2,500 of the 9,797 such families (seed 0), giving 5,578 asks (mean 6.2 words, mostly two per family). The knowledge base (86,942 passages, mean 56 words) contains the canonical passages, the five passages on each side of each canonical passage by passage id, and 60,000 passages drawn at random. MS MARCO passages were collected in blocks per original query, so id neighbours are topical distractors: their mean bge cosine similarity to the canonical passage is 0.72, against 0.42 for random passages. Because the full 3.4 GB collection could not be downloaded in the time available, ids were restricted to the first 4.3 million passages (about half the collection). Neighbours and random passages are unjudged, so, as with CQADupStack, some answers counted as wrong may be correct. The loop, conditions, grid and orders are those of version 2, with bge, MiniLM and BM25, for question, thread-copy and question-plus-thread entries.

Results. Table 10 and Figure 5 (right) show the same qualitative pattern as on CQADupStack, shifted towards safety. Base accuracy is higher (47.2% for bge, 44.4% for MiniLM, 17.7% for BM25). With question entries, write-back helps more at low acceptance rates (+2.19 points at $p_w = 0.1$ for bge) and still hurts at $p_w = 0.95$ (-1.43 for bge, -0.42 for MiniLM), with crossovers 0.58 [0.55, 0.61] (bge) and 0.78 [0.74, 0.86] (MiniLM), higher than the CQADupStack medians. The pre-simulation

Retriever	Entry content	Top-1	$\Delta@0.1$	$\Delta@0.3$	$\Delta@0.95$	p^*	self-share @0.95
bge	ask text	47.2	$+2.19 \pm 0.22$	$+1.06 \pm 0.22$	-1.43 ± 0.20	0.58 [0.55, 0.61]	54%
	ask + thread text		$+1.29 \pm 0.10$	$+1.00 \pm 0.12$	$+0.33 \pm 0.13$	>0.95	33%
	shown thread text		$+0.00 \pm 0.00$	$+0.00 \pm 0.00$	$+0.00 \pm 0.00$	>0.95	0%
MiniLM	ask text	44.4	$+1.94 \pm 0.14$	$+1.27 \pm 0.17$	-0.42 ± 0.23	0.78 [0.74, 0.86]	50%
	ask + thread text		$+0.78 \pm 0.06$	$+0.60 \pm 0.09$	$+0.13 \pm 0.09$	>0.95	29%
	shown thread text		$+0.00 \pm 0.00$	$+0.00 \pm 0.00$	$+0.00 \pm 0.00$	>0.95	0%
BM25	ask text	17.7	$+0.59 \pm 0.09$	$+0.43 \pm 0.06$	-0.07 ± 0.10	0.86 [0.69, 0.95]	21%
	ask + thread text		$+0.87 \pm 0.11$	$+0.61 \pm 0.10$	$+0.01 \pm 0.15$	>0.95	42%
	shown thread text		$+0.00 \pm 0.00$	$+0.00 \pm 0.00$	$+0.00 \pm 0.00$	>0.95	0%

Table 10: MS MARCO families: no-write-back top-1 accuracy (%), paired change in top-1 accuracy (points, 95% interval over ten orders) under naive write-back, crossover with 95% bootstrap interval, and share of top-1 answers from written-back entries in the second half of the stream at $p_w = 0.95$.

diagnostic of Section 4 explains the shift: under bge, an ask from a different family outscores the best passage for 24% of MS MARCO asks against 55% on CQADupStack (MiniLM 21% against 29%). Short queries are still closer to each other than to passages, but less often than short titles are to other titles. Question-plus-thread entries are non-negative at every tested p_w for both dense retrievers ($+0.33$ and $+0.13$ at 0.95) and keep about half or more of the gain at $p_w = 0.3$ ($+1.00$ against $+1.06$ for bge). Demotion makes question entries non-negative at $p_w = 0.95$ ($+0.41$ for bge). BM25 is not immune here: written-back entries supply 21% of top-1 answers at $p_w = 0.95$, give a gain of $+0.43$ points at $p_w = 0.3$, and the crossover is 0.86 [0.69, 0.95]; the change at $p_w = 0.95$ (-0.07) is not significant. The two version-3 baselines behave as on CQADupStack, with one difference. The cross-encoder filter ($q = 0.5$) does better here: for question entries it reduces the change at $p_w = 0.95$ to -0.41 (bge; crossover 0.71), $+0.49$ (MiniLM) and $+0.39$ (BM25), while keeping part of the gain at $p_w = 0.3$ ($+0.77$ for bge). The cross-encoder was trained on MS MARCO training data, which includes the query–passage judgements these families are built from, so this result overstates what an off-the-shelf filter would do on unseen data. Demotion is again at least as good ($+0.41$ and $+1.88$ for bge). The semantic cache with a loose threshold ($q = 0.5$) answers 24.0% of asks with hit precision 42%, below the base accuracy of 47.2% (bge), helps at $p_w = 0.3$ ($+0.63$) and hurts at $p_w = 0.95$ (-0.56 ; BM25 -0.98 , crossover 0.39); with a strict threshold ($q = 0.9$) it is close to inert (4.2% hits).

This corpus differs from CQADupStack in more than document type (queries are web searches rather than forum titles, families are smaller, distractors are partly random), so it does not isolate the effect of document type. It does show that the harm of question entries, the crossover, and the trade-off between question and question-plus-thread entries are not artefacts of a corpus made of questions.

12 Limitations

Simulated acceptance. User acceptance is the only simulated component, and it is the one that matters most. Real acceptance probably depends on the question, the answer and the user, and acceptance of wrong answers is likely correlated with how plausible they look [Joachims et al., 2007]. Version 2 adds one confidence-dependent acceptance model, which moves crossovers slightly down; real acceptance may depend on other things, such as answer length or user expertise. The plug-in model would still apply if the per-entry effects were measured under the real acceptance process, but the crossovers reported here would not. Every p^* in this paper is specific to a corpus, retriever, entry design and acceptance model; none should be read as a general threshold. **Incomplete**

ground truth. CQADupStack duplicate links are incomplete [Hoogeveen et al., 2015]: an answer we count as wrong may in fact solve the problem. This affects absolute accuracies and could make some “wrong” entries harmless, which would raise p^* . **Short questions and corpora.** CQADupStack asks are post titles and its documents are questions, which makes retrieval hard (base top-1 accuracy 9–35%) and favours question-to-question matches. The MS MARCO corpus has non-question documents and higher base accuracy, but it is built from about half the passage collection (download limit), its distractors are id neighbours and random passages rather than the full collection, and its queries are web searches rather than support tickets; it is a second data point, not a controlled test of document type. Entries containing a generated answer were not tested. **Simplified write-back.** Entries point to an existing thread; there is no generation step, so we do not measure errors introduced by an LLM writing the answer. BM25 statistics are not updated as entries are added. The probe bank is drawn from the stream itself. **Scope of the model.** The model is first order and is checked on the same simulated data it describes. The out-of-sample test splits orders, not forums or data, so it is a weak form of out-of-sample evidence. The confidence gate and footprint thresholds were set by simple rules, not tuned; tuned versions may do better. The cross-encoder filter uses a model trained on web queries and passages; a filter trained on in-domain data, per-entry cache thresholds [Schroeder et al., 2026] or evidence checks [Shah, 2026] were not tested. The filter and cache baselines were run on CQADupStack and MS MARCO but the confidence-dependent acceptance model was not combined with them.

13 Conclusion

On real repeat questions, writing accepted questions back into a dense retriever’s knowledge base helps only while users reject most wrong answers. This is mostly a property of what is stored: a stored question wins because it resembles the next question, and storing it with the thread text, or reranking with a cross-encoder, removes most of the loss together with most of the gain. The acceptance rate at which write-back starts to hurt varies from below 0.10 to about 0.86 on CQADupStack and is higher on MS MARCO query families; it is not predicted by base accuracy, but it is predicted reasonably well by a first-order formula once the per-entry effects of correct and wrong entries are measured. Reopening wrong answers reduces the loss more than a confidence gate, a footprint rule or a cross-encoder relevance filter, and serving stored questions as a similarity-threshold cache is harmful unless the threshold is so strict that the cache does almost nothing. The label-free monitors we tried add little to the acceptance rate. For practitioners: decide what an entry contains before deciding how to police it, let users reopen wrong answers and remove the corresponding entries, do not serve stored questions directly on similarity alone, and measure the acceptance rate of wrong answers and the per-entry effect of a wrong entry before turning on write-back.

Code and data availability

All data are public (CQADupStack via the BEIR/MTEB distribution on Hugging Face; MS MARCO passage ranking via the MTEB mirror). Code, logs and the scripts that generate every table and figure are at <https://github.com/pavansky/self-updating-retrieval>. Every number in the paper is produced by these scripts from saved runs. The version-3 additions (prep_v3.py, sim_v3.py, prep_het_v3.py, analyze_v3.py) take about two hours on a laptop, most of it cross-encoder scoring, embedding the MS MARCO passages and the iterated simulation.

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This work uses no employer or customer data. The retrieval setup follows TicketIQ, the author’s M.Tech dissertation system.

Changes from version 2

- Related work rewritten after a literature search (October 2026). It now discusses agent memory with self-written experience [Xiong et al., 2026b], self-improving and write-back RAG [Chinthala, 2025, Lu et al., 2026], query-based document expansion and its filtering [Scholer et al., 2004, Nogueira et al., 2019, Gospodinov et al., 2023], question retrieval [Jeon et al., 2005, Sakata et al., 2019], memory and cache poisoning [Dong et al., 2025, Chen et al., 2024b, Zhang et al., 2026b], implicit-feedback bias [Joachims et al., 2005, 2007] and degenerate feedback loops [Jiang et al., 2019]. The claim in version 2 that prior studies of such loops lack a human gate was removed; the novelty statement is narrowed to the human-gated, non-generative case, the crossover and its prediction, and the entry-design trade-off.
- New: a Doc2Query-- style cross-encoder filter and a semantic-cache baseline (Section 8); a second corpus built from MS MARCO query families (Section 11).
- Changed claims: BM25’s near-immunity is specific to ordinary write-back on CQADupStack; on MS MARCO, BM25 entries take over part of the answers, and served as a cache they are harmful. Version 2’s conclusion that demotion is the best of the safeguards tried still holds with the two new baselines. All version-2 numbers are unchanged.

Changes from version 1

- Related work now covers generated content in retrieval corpora [Chen et al., 2024a, Dai et al., 2024, Yu et al., 2026b, Druck and Smith, 2026] and case-based reasoning [Aamodt and Plaza, 1994, Smyth and Keane, 1995, Zhu and Yang, 1999]; the abstract and introduction describe the setting as the retrieval half of RAG.
- New: entry-content ablation and capture/steal diagnostics (Section 5); confidence-dependent acceptance (Section 6); confidence-gate and footprint baselines (Section 7); out-of-sample test of the plug-in estimate (Section 9); cross-encoder reranker on four forums.
- Changed claims: the harm reported in v1 is specific to storing question text; it largely disappears with question-plus-thread entries or a cross-encoder reranker. The plug-in estimate’s error is about 0.085 out of sample rather than 0.049 in sample. All v1 numbers for question entries are reproduced exactly by the v2 code.

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A Full safeguard results

Forum	Retriever	p_w	Naive	+ demotion	+ demotion + quarantine
android	bge	0.3	+0.54 $\pm$ 0.53	+1.79 $\pm$ 0.67	+2.46 $\pm$ 0.90
		0.95	-1.87 $\pm$ 0.65	-0.43 $\pm$ 0.67	-0.44 $\pm$ 0.72
	MiniLM	0.3	+0.54 $\pm$ 0.28	+0.92 $\pm$ 0.20	+0.84 $\pm$ 0.36
		0.95	-0.29 $\pm$ 0.24	+0.31 $\pm$ 0.23	+0.30 $\pm$ 0.26
english	bge	0.3	+1.25 $\pm$ 0.45	+1.95 $\pm$ 0.38	+2.16 $\pm$ 0.49
		0.95	-0.81 $\pm$ 0.44	+0.64 $\pm$ 0.36	+0.61 $\pm$ 0.37
	MiniLM	0.3	+1.84 $\pm$ 0.46	+2.83 $\pm$ 0.26	+2.85 $\pm$ 0.47
		0.95	-0.21 $\pm$ 0.42	+1.44 $\pm$ 0.28	+1.34 $\pm$ 0.26
gaming	bge	0.3	-0.44 $\pm$ 0.31	-0.05 $\pm$ 0.26	+0.35 $\pm$ 0.28
		0.95	-1.92 $\pm$ 0.20	-0.92 $\pm$ 0.22	-0.93 $\pm$ 0.23
	MiniLM	0.3	+0.15 $\pm$ 0.10	+0.39 $\pm$ 0.12	+0.67 $\pm$ 0.19
		0.95	-0.45 $\pm$ 0.23	+0.03 $\pm$ 0.20	+0.03 $\pm$ 0.21
gis	bge	0.3	-0.53 $\pm$ 0.32	+0.06 $\pm$ 0.27	+0.02 $\pm$ 0.27
		0.95	-2.23 $\pm$ 0.33	-1.08 $\pm$ 0.24	-1.11 $\pm$ 0.27
	MiniLM	0.3	-0.18 $\pm$ 0.18	-0.07 $\pm$ 0.13	-0.08 $\pm$ 0.13
		0.95	-0.55 $\pm$ 0.18	-0.30 $\pm$ 0.15	-0.30 $\pm$ 0.15
mathematica	bge	0.3	-0.87 $\pm$ 0.24	-0.30 $\pm$ 0.30	-0.23 $\pm$ 0.30
		0.95	-2.58 $\pm$ 0.30	-1.41 $\pm$ 0.31	-1.41 $\pm$ 0.31
	MiniLM	0.3	-0.19 $\pm$ 0.29	+0.40 $\pm$ 0.26	+0.29 $\pm$ 0.29
		0.95	-1.15 $\pm$ 0.27	-0.42 $\pm$ 0.22	-0.43 $\pm$ 0.21
physics	bge	0.3	-0.03 $\pm$ 0.28	+0.56 $\pm$ 0.34	+0.59 $\pm$ 0.32
		0.95	-1.70 $\pm$ 0.39	-0.72 $\pm$ 0.32	-0.73 $\pm$ 0.32
	MiniLM	0.3	+0.32 $\pm$ 0.27	+0.64 $\pm$ 0.21	+0.72 $\pm$ 0.16
		0.95	-0.74 $\pm$ 0.29	+0.04 $\pm$ 0.32	+0.03 $\pm$ 0.32
programmers	bge	0.3	+0.27 $\pm$ 0.29	+0.88 $\pm$ 0.25	+1.00 $\pm$ 0.15
		0.95	-1.73 $\pm$ 0.28	-0.42 $\pm$ 0.27	-0.41 $\pm$ 0.27
	MiniLM	0.3	+0.31 $\pm$ 0.31	+0.60 $\pm$ 0.27	+0.51 $\pm$ 0.28
		0.95	-0.48 $\pm$ 0.24	+0.07 $\pm$ 0.31	+0.07 $\pm$ 0.31
stats	bge	0.3	-0.82 $\pm$ 0.24	-0.35 $\pm$ 0.15	-0.19 $\pm$ 0.17
		0.95	-1.73 $\pm$ 0.22	-1.18 $\pm$ 0.24	-1.18 $\pm$ 0.24
	MiniLM	0.3	-0.33 $\pm$ 0.26	-0.19 $\pm$ 0.24	-0.22 $\pm$ 0.21
		0.95	-1.11 $\pm$ 0.21	-0.56 $\pm$ 0.30	-0.57 $\pm$ 0.29
tex	bge	0.3	-0.81 $\pm$ 0.17	-0.25 $\pm$ 0.24	+0.13 $\pm$ 0.38
		0.95	-2.62 $\pm$ 0.18	-1.49 $\pm$ 0.23	-1.49 $\pm$ 0.24
	MiniLM	0.3	-0.21 $\pm$ 0.15	+0.13 $\pm$ 0.15	+0.36 $\pm$ 0.17
		0.95	-1.03 $\pm$ 0.11	-0.42 $\pm$ 0.16	-0.41 $\pm$ 0.16
unix	bge	0.3	-0.32 $\pm$ 0.25	+0.58 $\pm$ 0.39	+0.73 $\pm$ 0.36
		0.95	-2.68 $\pm$ 0.57	-1.03 $\pm$ 0.51	-1.03 $\pm$ 0.51
	MiniLM	0.3	+0.62 $\pm$ 0.24	+0.73 $\pm$ 0.23	+0.82 $\pm$ 0.24
		0.95	-0.47 $\pm$ 0.24	+0.27 $\pm$ 0.19	+0.21 $\pm$ 0.20
webmasters	bge	0.3	-0.45 $\pm$ 0.35	+0.04 $\pm$ 0.33	+0.23 $\pm$ 0.35
		0.95	-1.79 $\pm$ 0.28	-0.93 $\pm$ 0.41	-0.92 $\pm$ 0.41
	MiniLM	0.3	+0.51 $\pm$ 0.35	+0.85 $\pm$ 0.35	+1.20 $\pm$ 0.44
		0.95	-0.32 $\pm$ 0.26	+0.23 $\pm$ 0.36	+0.24 $\pm$ 0.35
wordpress	bge	0.3	-0.27 $\pm$ 0.16	+0.01 $\pm$ 0.11	+0.17 $\pm$ 0.14
		0.95	-0.99 $\pm$ 0.34	-0.43 $\pm$ 0.41	-0.43 $\pm$ 0.41
	MiniLM	0.3	+0.15 $\pm$ 0.28	+0.19 $\pm$ 0.18	+0.15 $\pm$ 0.17
		0.95	-0.11 $\pm$ 0.24	+0.11 $\pm$ 0.23	+0.11 $\pm$ 0.23

Table 11: Paired change in top-1 accuracy (points, 95% interval over ten orders) relative to no write-back, for the dense retrievers.

Forum	Retriever	ask text (v1)	shown thread	ask + thread	ask + thread, +demotion
android	bge	0.47 [0.37, 0.54]	>0.95	0.77 [0.49, 0.95]	>0.95
	MiniLM	0.67 [0.55, 0.78]	>0.95	>0.95	>0.95
	BM25	>0.95	>0.95	>0.95	>0.95
english	bge	0.63 [0.50, 0.75]	>0.95	0.80 [0.67, 0.90]	>0.95
	MiniLM	0.86 [0.77, 0.95]	>0.95	>0.95	>0.95
	BM25	>0.95	>0.95	>0.95	>0.95
gaming	bge	0.15 [0.10, 0.23]	>0.95	0.69 [0.19, 0.95]	>0.95
	MiniLM	0.45 [0.36, 0.57]	>0.95	0.91 [0.59, 0.95]	>0.95
	BM25	<0.10	>0.95	0.58 [0.10, 0.95]	0.54 [0.24, 0.95]
gis	bge	0.15 [0.10, 0.22]	>0.95	0.69 [0.20, 0.95]	>0.95
	MiniLM	<0.10	>0.95	0.36 [0.10, 0.79]	>0.95
	BM25	>0.95	>0.95	>0.95	0.10 [0.10, 0.95]
mathematica	bge	<0.10	>0.95	0.70 [0.36, 0.95]	>0.95
	MiniLM	0.23 [0.16, 0.31]	>0.95	<0.10	0.17 [0.10, 0.45]
	BM25	>0.95	>0.95	<0.10	0.10 [0.10, 0.31]
physics	bge	0.29 [0.19, 0.34]	>0.95	0.69 [0.44, 0.84]	0.94 [0.69, 0.95]
	MiniLM	0.40 [0.34, 0.48]	>0.95	0.65 [0.43, 0.85]	>0.95
	BM25	>0.95	>0.95	0.34 [0.17, 0.43]	0.49 [0.37, 0.79]
programmers	bge	0.34 [0.30, 0.38]	>0.95	0.74 [0.45, 0.95]	>0.95
	MiniLM	0.57 [0.33, 0.65]	>0.95	0.70 [0.35, 0.92]	>0.95
	BM25	>0.95	>0.95	>0.95	>0.95
stats	bge	<0.10	>0.95	>0.95	>0.95
	MiniLM	<0.10	>0.95	0.55 [0.44, 0.66]	>0.95
	BM25	>0.95	>0.95	>0.95	>0.95
tex	bge	0.11 [0.10, 0.14]	>0.95	0.89 [0.66, 0.95]	>0.95
	MiniLM	0.22 [0.16, 0.27]	>0.95	>0.95	>0.95
	BM25	0.10 [0.10, 0.40]	>0.95	0.20 [0.10, 0.80]	0.30 [0.10, 0.95]
unix	bge	0.26 [0.24, 0.29]	>0.95	0.81 [0.58, 0.95]	>0.95
	MiniLM	0.63 [0.52, 0.74]	>0.95	>0.95	>0.95
	BM25	>0.95	>0.95	>0.95	>0.95
webmasters	bge	0.14 [0.12, 0.19]	>0.95	0.32 [0.14, 0.66]	0.45 [0.23, 0.77]
	MiniLM	0.77 [0.56, 0.92]	>0.95	0.90 [0.26, 0.95]	>0.95
	BM25	<0.10	>0.95	<0.10	<0.10
wordpress	bge	0.15 [0.10, 0.20]	>0.95	>0.95	>0.95
	MiniLM	0.39 [0.18, 0.95]	>0.95	<0.10	<0.10
	BM25	0.35 [0.10, 0.90]	>0.95	>0.95	>0.95

Table 12: Crossover p^* (95% bootstrap interval) by entry content, naive write-back unless stated.

Forum	Retriever	p^* constant	p^* difficulty-dep.	$\Delta@0.95$ constant	$\Delta@0.95$ difficulty-dep.
android	bge	0.47 [0.37, 0.54]	0.37 [0.32, 0.48]	-1.87 ± 0.65	-1.90 ± 0.63
	MiniLM	0.67 [0.55, 0.78]	0.59 [0.44, 0.71]	-0.29 ± 0.24	-0.31 ± 0.23
	BM25	>0.95	>0.95	$+0.03 \pm 0.10$	$+0.03 \pm 0.10$
english	bge	0.63 [0.50, 0.75]	0.42 [0.34, 0.55]	-0.81 ± 0.44	-0.98 ± 0.53
	MiniLM	0.86 [0.77, 0.95]	0.72 [0.63, 0.95]	-0.21 ± 0.42	-0.31 ± 0.35
	BM25	>0.95	>0.95	$+0.07 \pm 0.04$	$+0.06 \pm 0.04$
gaming	bge	0.15 [0.10, 0.23]	0.12 [0.10, 0.16]	-1.92 ± 0.20	-1.98 ± 0.19
	MiniLM	0.45 [0.36, 0.57]	0.34 [0.26, 0.50]	-0.45 ± 0.23	-0.47 ± 0.23
	BM25	<0.10	<0.10	-0.11 ± 0.08	-0.11 ± 0.08
gis	bge	0.15 [0.10, 0.22]	<0.10	-2.23 ± 0.33	-2.24 ± 0.34
	MiniLM	<0.10	<0.10	-0.55 ± 0.18	-0.56 ± 0.18
	BM25	>0.95	>0.95	$+0.00 \pm 0.00$	$+0.00 \pm 0.00$
mathematica	bge	<0.10	<0.10	-2.58 ± 0.30	-2.56 ± 0.29
	MiniLM	0.23 [0.16, 0.31]	0.18 [0.12, 0.24]	-1.15 ± 0.27	-1.18 ± 0.27
	BM25	>0.95	>0.95	$+0.00 \pm 0.00$	$+0.00 \pm 0.00$
physics	bge	0.29 [0.19, 0.34]	0.24 [0.18, 0.27]	-1.70 ± 0.39	-1.72 ± 0.32
	MiniLM	0.40 [0.34, 0.48]	0.33 [0.23, 0.44]	-0.74 ± 0.29	-0.76 ± 0.27
	BM25	>0.95	>0.95	$+0.00 \pm 0.00$	$+0.00 \pm 0.00$
programmers	bge	0.34 [0.30, 0.38]	0.21 [0.18, 0.28]	-1.73 ± 0.28	-1.79 ± 0.30
	MiniLM	0.57 [0.33, 0.65]	0.29 [0.20, 0.44]	-0.48 ± 0.24	-0.54 ± 0.26
	BM25	>0.95	>0.95	$+0.01 \pm 0.02$	$+0.01 \pm 0.02$
stats	bge	<0.10	<0.10	-1.73 ± 0.22	-1.75 ± 0.22
	MiniLM	<0.10	<0.10	-1.11 ± 0.21	-1.12 ± 0.20
	BM25	>0.95	>0.95	$+0.12 \pm 0.11$	$+0.12 \pm 0.11$
tex	bge	0.11 [0.10, 0.14]	<0.10	-2.62 ± 0.18	-2.64 ± 0.16
	MiniLM	0.22 [0.16, 0.27]	0.16 [0.12, 0.21]	-1.03 ± 0.11	-1.04 ± 0.11
	BM25	0.10 [0.10, 0.40]	0.10 [0.10, 0.20]	-0.01 ± 0.01	-0.01 ± 0.01
unix	bge	0.26 [0.24, 0.29]	0.20 [0.16, 0.23]	-2.68 ± 0.57	-2.77 ± 0.55
	MiniLM	0.63 [0.52, 0.74]	0.47 [0.41, 0.67]	-0.47 ± 0.24	-0.52 ± 0.25
	BM25	>0.95	>0.95	$+0.04 \pm 0.05$	$+0.04 \pm 0.05$
webmasters	bge	0.14 [0.12, 0.19]	0.13 [0.10, 0.19]	-1.79 ± 0.28	-1.78 ± 0.28
	MiniLM	0.77 [0.56, 0.92]	0.60 [0.32, 0.95]	-0.32 ± 0.26	-0.27 ± 0.32
	BM25	<0.10	<0.10	-0.08 ± 0.03	-0.08 ± 0.03
wordpress	bge	0.15 [0.10, 0.20]	0.10 [0.10, 0.16]	-0.99 ± 0.34	-1.01 ± 0.31
	MiniLM	0.39 [0.18, 0.95]	0.59 [0.27, 0.95]	-0.11 ± 0.24	-0.17 ± 0.26
	BM25	0.35 [0.10, 0.90]	<0.10	-0.07 ± 0.05	-0.07 ± 0.05

Table 13: Crossover and change at $p_w = 0.95$ under constant and confidence-dependent acceptance, question entries.